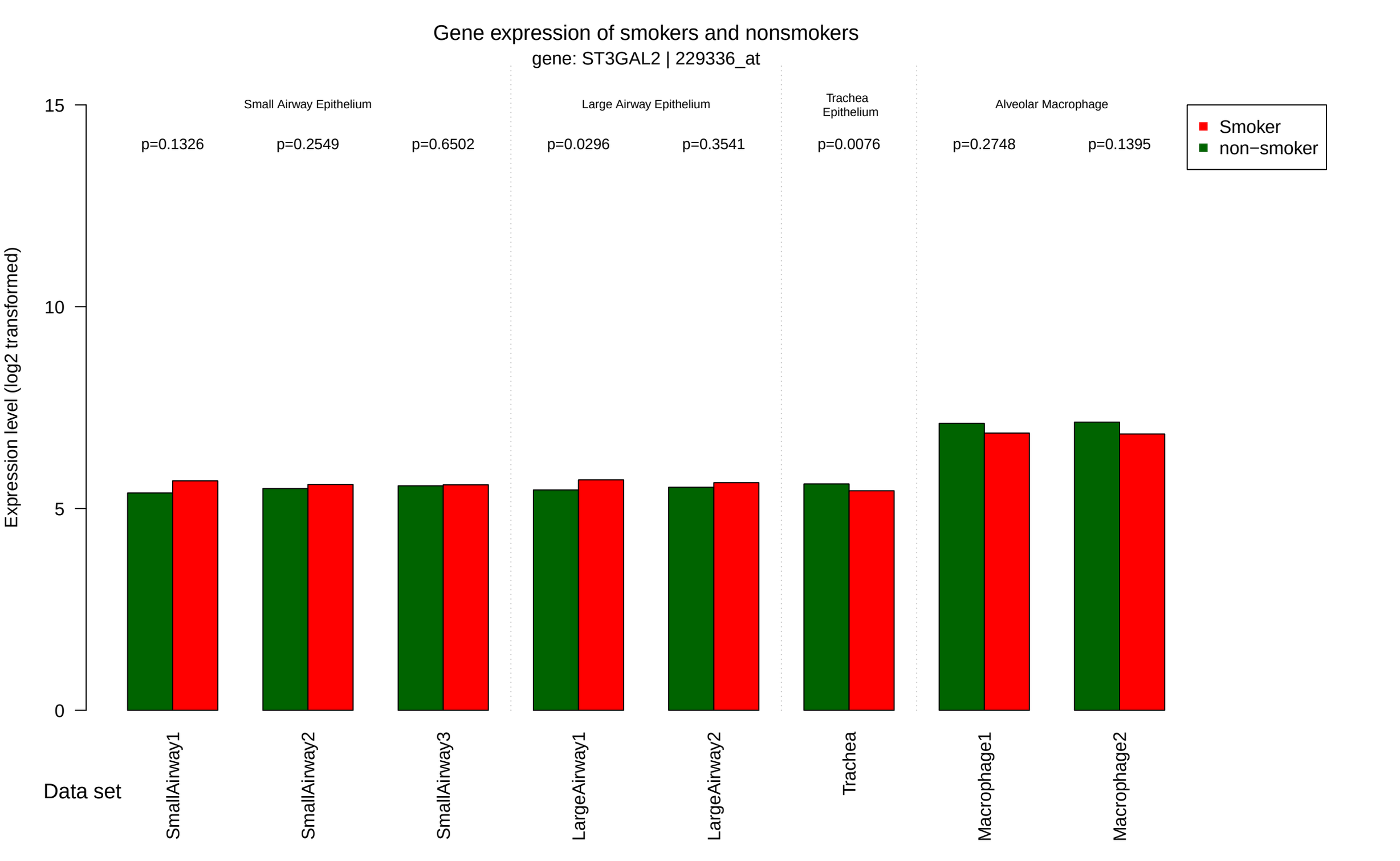
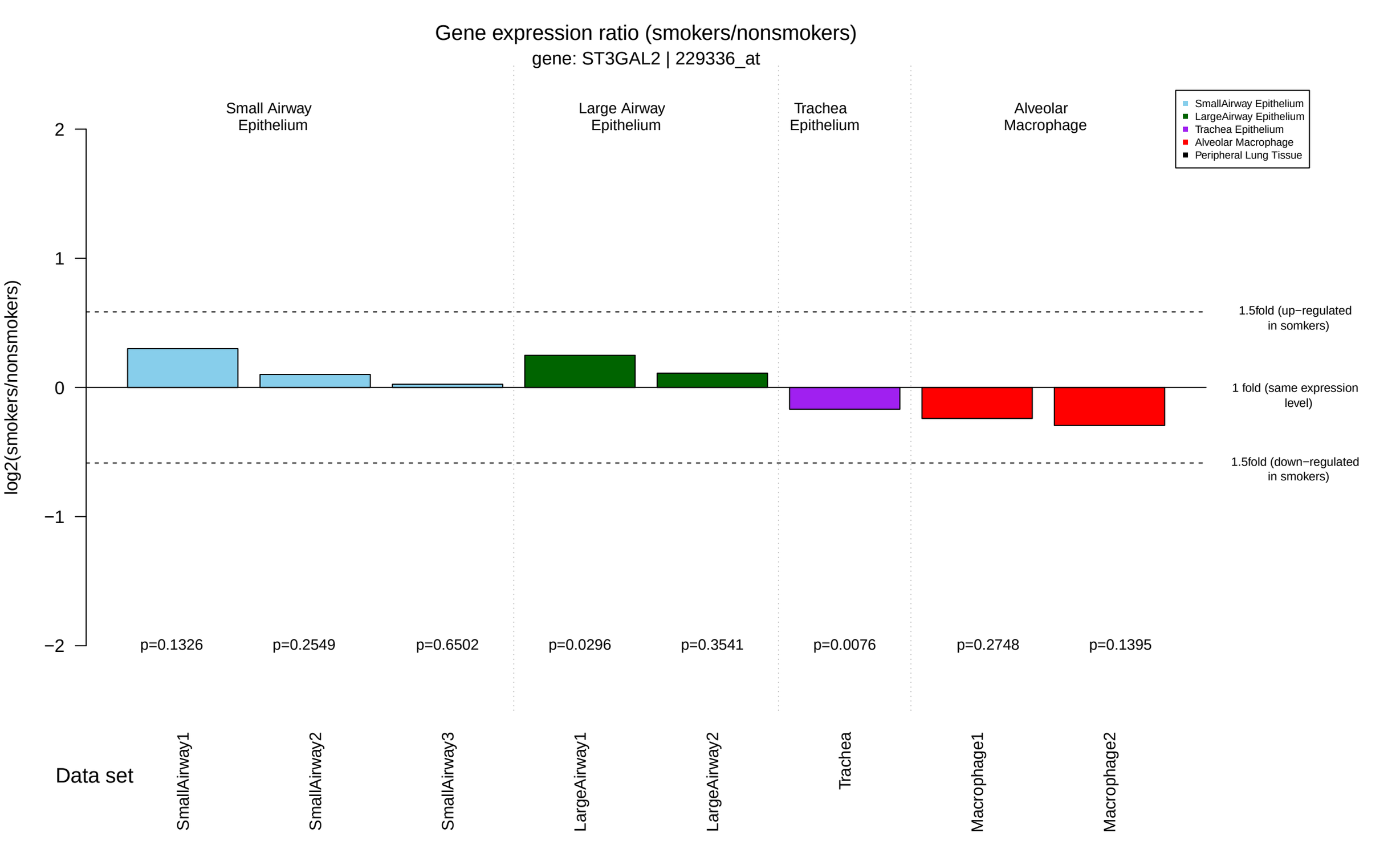
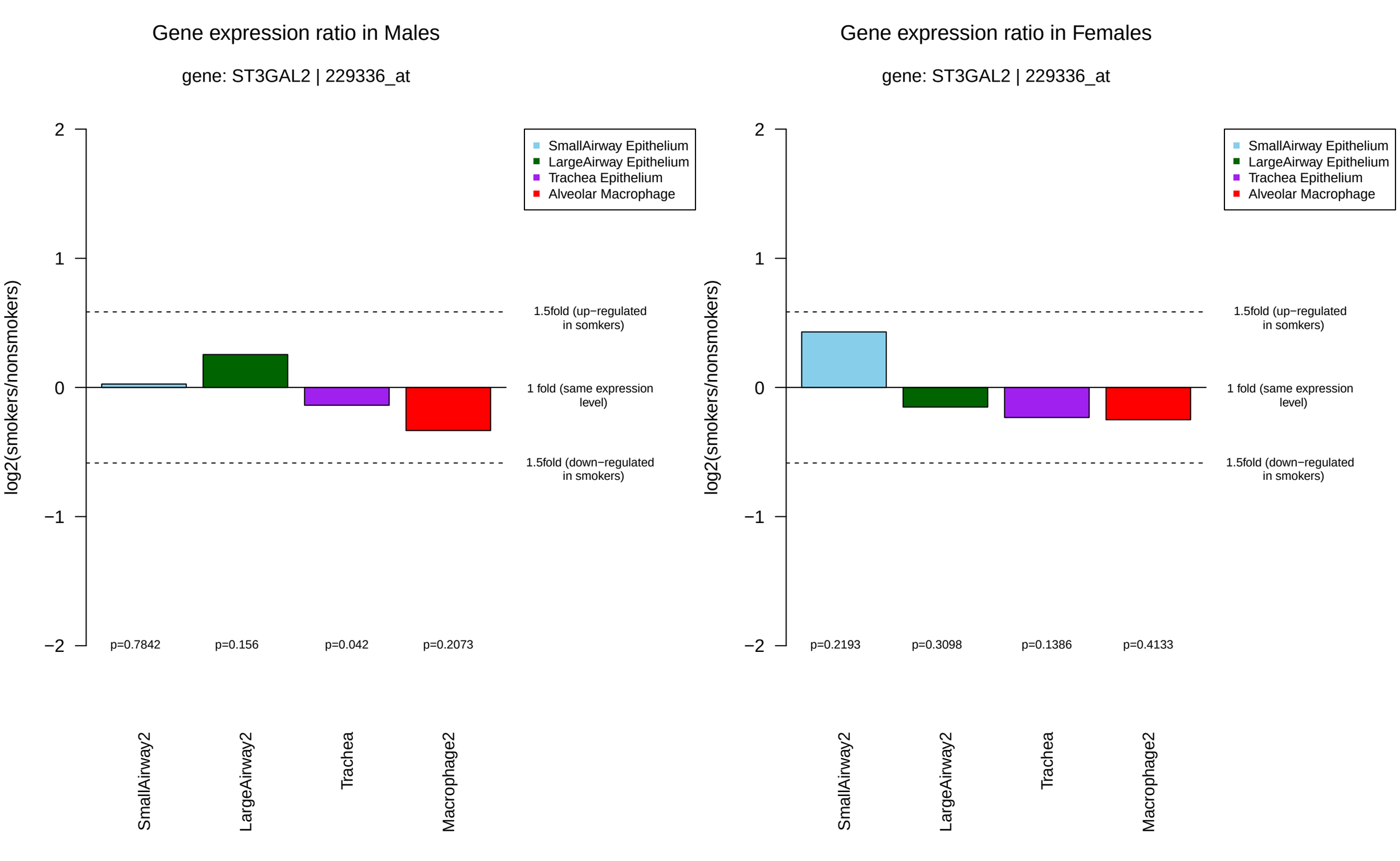
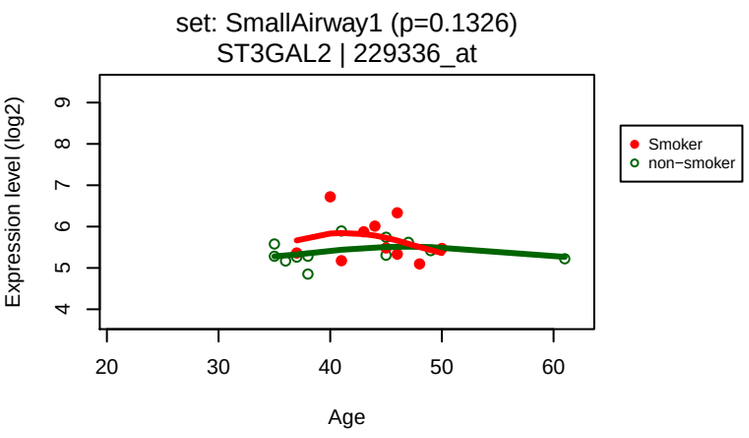
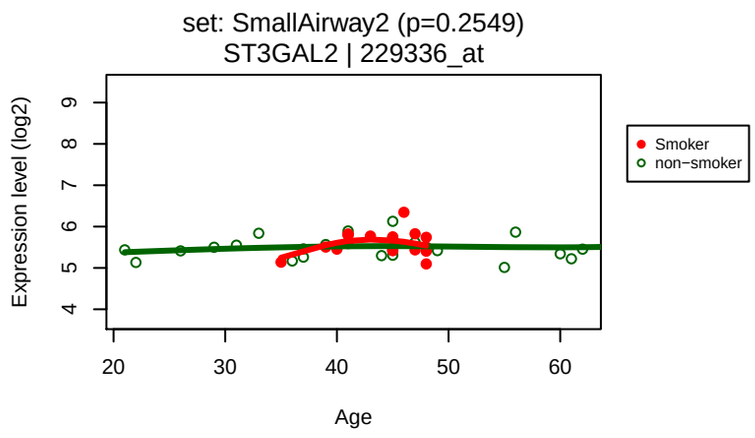
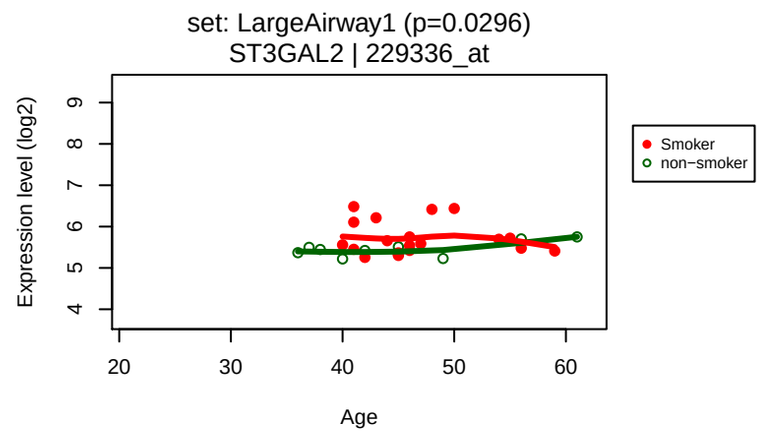

Gene expression of smokers and nonsmokers
gene: ST3GAL2 | 229336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1326
p=0.2549
p=0.6502
p=0.0296
p=0.3541
p=0.0076
p=0.2748
p=0.1395
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

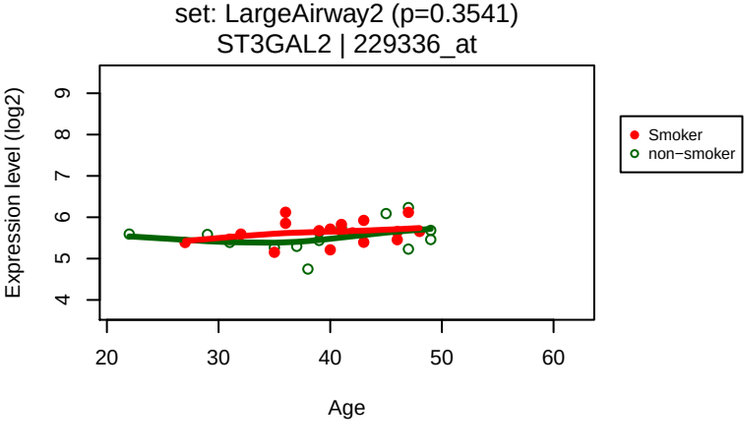

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL2 | 229336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1326
p=0.2549
p=0.6502
p=0.0296
p=0.3541
p=0.0076
p=0.2748
p=0.1395
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

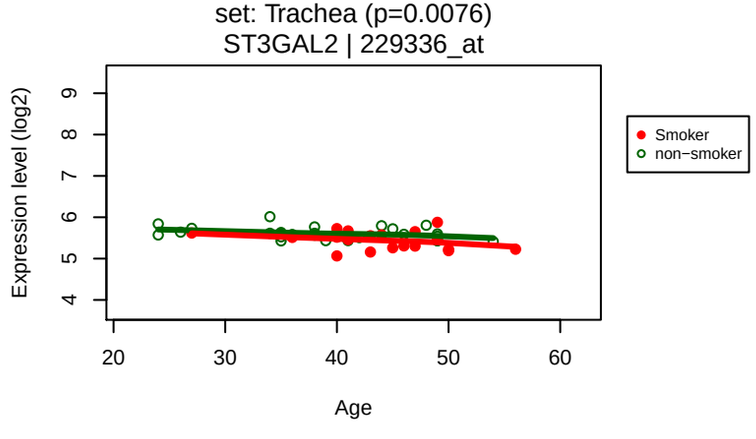

Gene expression ratio in Males
gene: ST3GAL2 | 229336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7842
p=0.156
p=0.042
p=0.2073
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL2 | 229336_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2193
p=0.3098
p=0.1386
p=0.4133
SmallAirway2
LargeAirway2
Trachea
Macrophage2

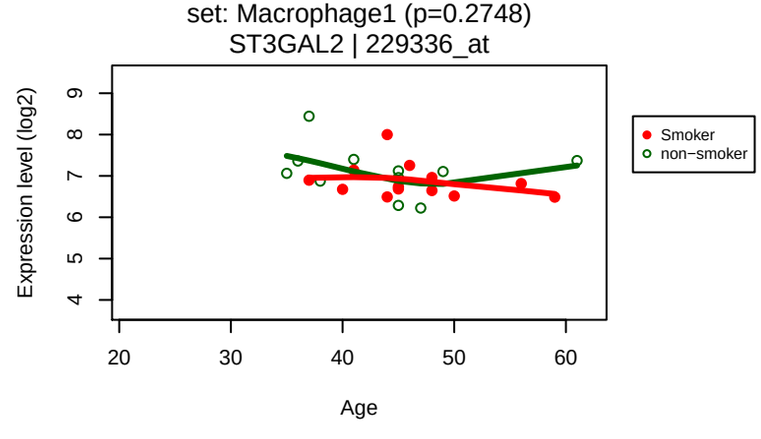

set: SmallAirway1 (p=0.1326)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker

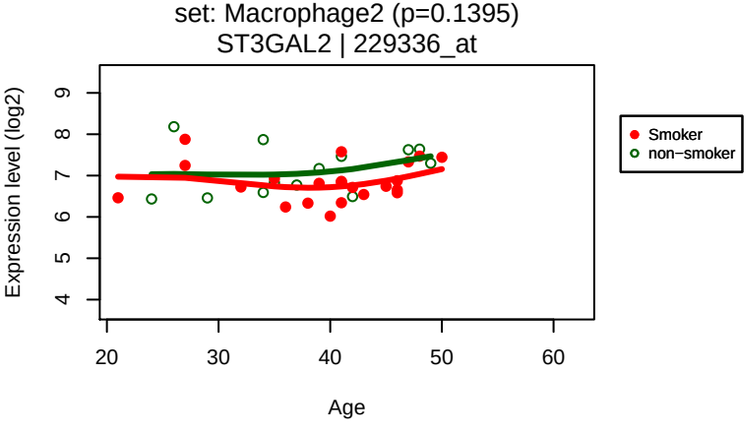

set: SmallAirway2 (p=0.2549)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker

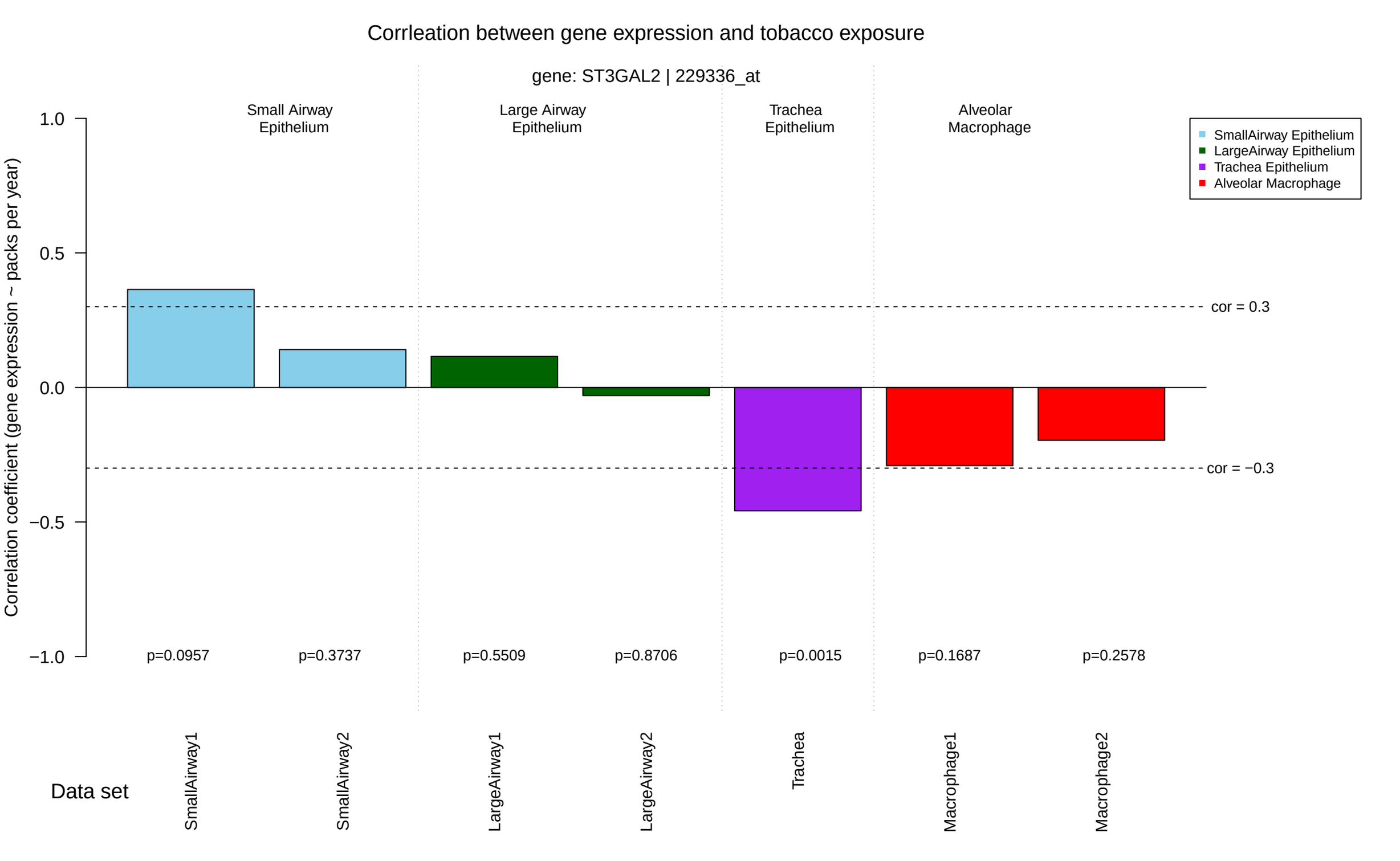

set: LargeAirway1 (p=0.0296)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3541)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0076)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2748)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1395)
ST3GAL2 | 229336_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL2 | 229336_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0957
p=0.3737
p=0.5509
p=0.8706
p=0.0015
p=0.1687
p=0.2578
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2